CHEST

Clinical Characteristics, Treatment Persistence, and Outcomes Among Patients With COPD Treated With Single- or Multiple-Inhaler Triple Therapy: A Retrospective Analysis in Spain

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BACKGROUND: COPD is a leading cause of death and disability. COPD therapy goals include reducing exacerbations and improving symptom control. Single-inhaler triple therapy (SITT) or multiple-inhaler triple therapy (MITT) is indicated for patients with frequent exacerbations despite bronchodilator therapy. No available evidence compares SITT vs MITT in Spain in terms of treatment persistence, exacerbations, and other outcomes.

RESEARCH QUESTION: Do COPD patients in Spain initiating SITT vs MITT have improved persistence, exacerbations, and health care resource utilization?

STUDY DESIGN AND METHODS: This real-world, observational, retrospective cohort study analyzed electronic health records in the Spanish National Healthcare System BIG-PAC database to identify COPD patients aged ≥ 40 years initiating SITT or MITT (using two or three inhalers) between June 1, 2018 and December 31, 2019. Comparative data on persistence (allowing up to 60 days without prescription refill), exacerbation rates, and health care resource utilization and costs during 12-month follow-up were analyzed. Multivariate adjusted analyses were performed.

RESULTS: Eligible patients (N = 4,625) initiating SITT (n = 1,011) or MITT (n = 3,614) had a mean age of 70.9 years; most were male (73.9%) with mainly moderate (62.0%) or severe (26.5%) airflow limitation. Between-cohort baseline characteristics were similar. At 12-month follow-up, SITT patients had higher persistence (hazard ratio [HR] = 1.37; 95% CI = 1.22-1.53; $P < .001$), reduced risk of exacerbations (HR = 0.68; 95% CI = 0.61-0.77; $P = .001$), and lower all-cause mortality risk (HR = 0.67; 95% CI = 0.63-0.71, $P = .027$), compared with MITT patients. SITT was associated with significantly reduced health care resource use (mean annual cost savings: €403 vs MITT). For both SITT and MITT, persistence was associated with improved exacerbation rates vs nonpersistence, and substantial adjusted mean annual cost savings (€2,115 and €2,700, respectively).

INTERPRETATION: Patients initiating SITT had a clinically relevant improvement in persistence leading to reductions in mortality, incidence of exacerbations, and health care resource use with consequent mean cost savings. CHEST 2022; 162(5):1017-1029

KEY WORDS: COPD; fixed-dose combinations; inhaled corticosteroids (ICS); long-acting β2-agonists (LABA); long-acting muscarinic antagonists (LAMA); multiple-inhaler triple therapy (MITT); persistence; single-inhaler triple therapy (SITT)

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Take-home Points

Study Question: Do patients with COPD in Spain initiating single-inhaler triple therapy (SITT) vs multiple-inhaler triple therapy (MITT) have improved persistence, exacerbations, and health care resource utilization?

Results: At 12-month follow-up, compared with MITT patients, SITT patients had higher persistence, a reduced risk of exacerbations, a lower risk of mortality, and reduced health care resource use with consequent mean cost savings.

Interpretation: Patients initiating SITT had a clinically relevant improvement in persistence, leading to reductions in mortality and in the incidence of exacerbations, compared with patients initiating MITT. SITT was associated with significant reductions in health care resource use with consequent cost savings.

COPD is the third leading cause of death globally, accounting for 3.23 million deaths in 2019, with more than 80% occurring in low- and middle-income countries.[1,2] In Spain, COPD is the fourth leading cause of death, accounting for nearly 7% of all deaths nationally.[3] COPD is characterized by persistent respiratory symptoms such as dyspnea, cough or the production of sputum, and restricted airflow.[4] A feature of COPD is the appearance of episodes of worsening symptoms—exacerbations that lead to substantial morbidity and mortality.[5] Current Spanish and international COPD guidelines recommend short- and long-acting bronchodilators for symptomatic relief and maintenance of COPD.[4,6] These include the anticholinergic classes of short-acting muscarinic antagonists and long-acting muscarinic antagonists (LAMA), and short-acting β2-agonists (SABA) and long-acting β2-agonists (LABA). The addition of inhaled corticosteroids (ICS) is indicated for patients with frequent exacerbations despite optimal bronchodilator therapy, particularly in patients with high blood eosinophils.[4]

Data from multiple clinical trials support the use of single-inhaler triple combination therapy compared with dual ICS/LABA[7-14] or LABA/LAMA therapy.[11-15] Triple combination therapy administered by a single inhaler was effective in reducing the risk of all-cause death and moderate or severe exacerbations compared with single-inhaler dual therapy.[12,13,16,17] In randomized controlled trials and real-world evidence studies, single-inhaler triple therapy (SITT) delivering ICS/LABA/LAMA was non-inferior[18,19] or superior[20] to multiple-inhaler triple therapy (MITT), administered via two inhalers, with respect to lung function improvement[18-20] and health status improvement.[19,20] Single (or fixed-dose) combination inhalers offer advantages to COPD patients by simplifying complex inhaler regimens,[21] and single-inhaler delivery of dual or triple therapy in COPD improves adherence and persistence compared with delivery using multiple inhalers.[22,23] However, there is little evidence regarding the effectiveness of SITT vs MITT in terms of exacerbation prevention and patient persistence, as well as economic impact.

We conducted this real-world evidence study to compare the effectiveness of SITT vs MITT in treatment persistence, exacerbation prevention, and economic consequences in a Spanish COPD population.

Study Design and Methods

Objectives

The main objective was based on the hypothesis that SITT may improve treatment persistence compared with MITT, potentially leading to a lower incidence of exacerbations and lower health care resource utilization. Thus, the study aimed to compare persistence between the SITT and MITT cohorts as the primary objective; and secondarily, the incidence of moderate/severe exacerbations and the economic consequences (health care resource utilization) between treatment cohorts. Another objective of the study was to describe and compare the clinical and demographic characteristics of the two COPD cohorts initiating SITT or MITT using two or three inhalers.

Study Design

The study was a real-world, observational, retrospective cohort design comprising longitudinal analyses of electronic health records.

ABBREVIATIONS: HR = hazard ratio; ICD-10 = International Classification of Diseases 10; ICS = inhaled corticosteroids; LABA = long-acting β2 agonists; LAMA = long-acting anticholinergics; MITT = multiple inhaler triple therapy; SABA = short-acting β2 agonists; SITT = single-inhaler triple therapy

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Study Population

COPD patients (defined as those with a diagnostic code of COPD and a postbronchodilator $FEV_1/FVC < 0.7$, according to COPD guidelines), aged 40 years or older, starting triple therapy (LAMA, LABA, ICS) between June 1, 2018, and December 31, 2019, with either SITT or MITT (using two or three inhalers), were included in the study. A minimum patient history of 12 months before the index date (MITT or SITT initiation) was required. Patients were followed up until treatment discontinuation or for a maximum of 12 months.

Exclusion criteria were individuals inactive in the database, individuals moving or relocating to areas outside the scope of the BIG-PAC database, patients permanently institutionalized, and patients with reduced expected survival (ie, those with terminal disease or dialysis). Also, patients with inconsistent data after manual review (eg, FEV1 not available) were excluded.

Data Source

Data were obtained from the BIG-PAC database (Atrys Health SA) following Collaborative Agreements with Health Authorities in seven Autonomous Communities across Spain. The BIG-PAC database is certified by the Spanish Agency of Medicines and Medical Devices and has acceptable representativeness of the Spanish population.[24] The database contains anonymized computerized records of approximately 1.9 million patients in the Spanish National Healthcare System (Instituto Nacional de la Salud) and includes integrated data from primary care centers and public hospitals, including details of general practitioner visits, emergency care, pharmacy and prescription data (verified daily dose record, time interval, and duration of each treatment administered), hospital admissions, sick leave, disability, and deaths. Data are loaded into the BIG-PAC database on a monthly basis. The BIG-PAC database complies with the 2018 Spanish Data Protection and Digital Rights Act.

Study Variables

The following data were retrieved from electronic health records: demographic data, including age and sex; baseline clinical data, which included the total number of exacerbations and severe exacerbations during the previous year, time since diagnosis, comorbidities (defined by International Classification of Diseases 10 [ICD-10] diagnostic codes), smoking status (defined by ICD-10 codes and collected through a specific internal protocol at each primary care site), BMI, lung function (spirometry data were registered manually or automatically in specific structured fields of the electronic health records), eosinophil count and concomitant medications; persistence rates at 6 months and 12 months, and mean persistence (in days); the frequency and rate of moderate/severe exacerbations during the observation period, and stratified by severity; longitudinal health care resource utilization data comprising COPD-related hospitalizations and length of stay, visits to general practitioner, specialists (eg, pneumologist), and emergency room, all-cause mortality, and procedures and medications. The types of exacerbations retrieved during the observation period, ie, in the 12 months before and after the index date, were moderate and severe exacerbations; because of the nature of the study, relying on data available in health electronic records, mild exacerbations could not be retrieved.

Definitions

A moderate exacerbation was defined as a claim(s) of oral corticosteroids (Anatomical Therapeutic Chemical code H02AB) or respiratory antibiotics (Anatomical Therapeutic Chemical code J01AA, J01CA), both prescribed for COPD and with no concurrent acute nonpulmonary infection (for antibiotics). A severe exacerbation was defined as a hospital admission with ICD-10 clinical modification (ICD-10-CM) code J44, or an emergency visit (ICD-10-CM codes J44.0/J44.1) in an outpatient hospital setting. Only ICD-10 codes specific for COPD were included (J40–43 codes for bronchitis and emphysema were not included). Physician diagnoses were not included in the study.

Persistence was defined as the time from the index date (MITT or SITT initiation) to treatment discontinuation. Discontinuation was defined as more than 60 days without a prescription refill of any MITT component or SITT. Persistence rates are the proportion of patients achieving persistence at 6 and 12 months, as defined previously.

Statistical Analysis

Data were retrieved from the database using customized SQL scripts and were validated by frequency distribution analysis and careful review to identify possible recording or coding errors.

Qualitative data were summarized by absolute and relative frequencies and quantitative data by mean, SD, median, percentiles and 95% CIs calculated where applicable. Cohorts were compared using analysis of variance, which was used as a generalization of Student's t test for comparison of means, χ^2 tests for independent groups, and the nonparametric Mann-Whitney U test (Wilcoxon sum of ranks) for quantitative variables that were not normally distributed.

Treatment persistence and duration were analyzed using Kaplan-Meier survival analysis, with the log-rank test used to statistically compare both groups. Percentage results were estimated, equivalent to one per 100 person-years (incidence rate; accumulated risk). Data were censored in the absence of an event (MITT or SITT discontinuation). Cox proportional risk regression was used to analyze persistence rates up to 12 months and adjusted for covariates: age, sex, time from diagnosis, FEV_1, Charlson Comorbidity Index, and previous exacerbations. Sensitivity analyses were performed for persistence using > 90 days without a prescription refill.

COPD-related health care costs were estimated using an analysis of covariance generalized linear model adjusted for covariates (age, sex, BMI, smoking status, time from diagnosis, FEV_1, eosinophil count, heart failure, renal failure, and Charlson Comorbidity Index). Contrasts were based on pairwise comparisons between estimated marginal means using bootstrap sampling simulation (1,000 simulation samples) and with a Bonferroni correction for multiple testing.

Results

The study population was selected as summarized in Figure 1. Of approximately 1.9 million electronic health records in the database, 46,476 patients were diagnosed with COPD, of whom 4,894 (10.5%) initiated triple therapy. Eligible patients (N = 4,625) initiated SITT (n = 1,011; 21.9%) or MITT (n = 3,614; 78.1%).

Baseline characteristics of the study population are summarized in Table 1. Subjects had a mean (SD) age of 70.9 (14.3) years, and most were male (73.9%). The most common comorbidities were arterial hypertension (61.9%), dyslipidemia (51.1%), and obesity (28.1%). Most subjects had moderate (62.0%) or severe (26.5%) airflow limitation and with a mean (SD) time since

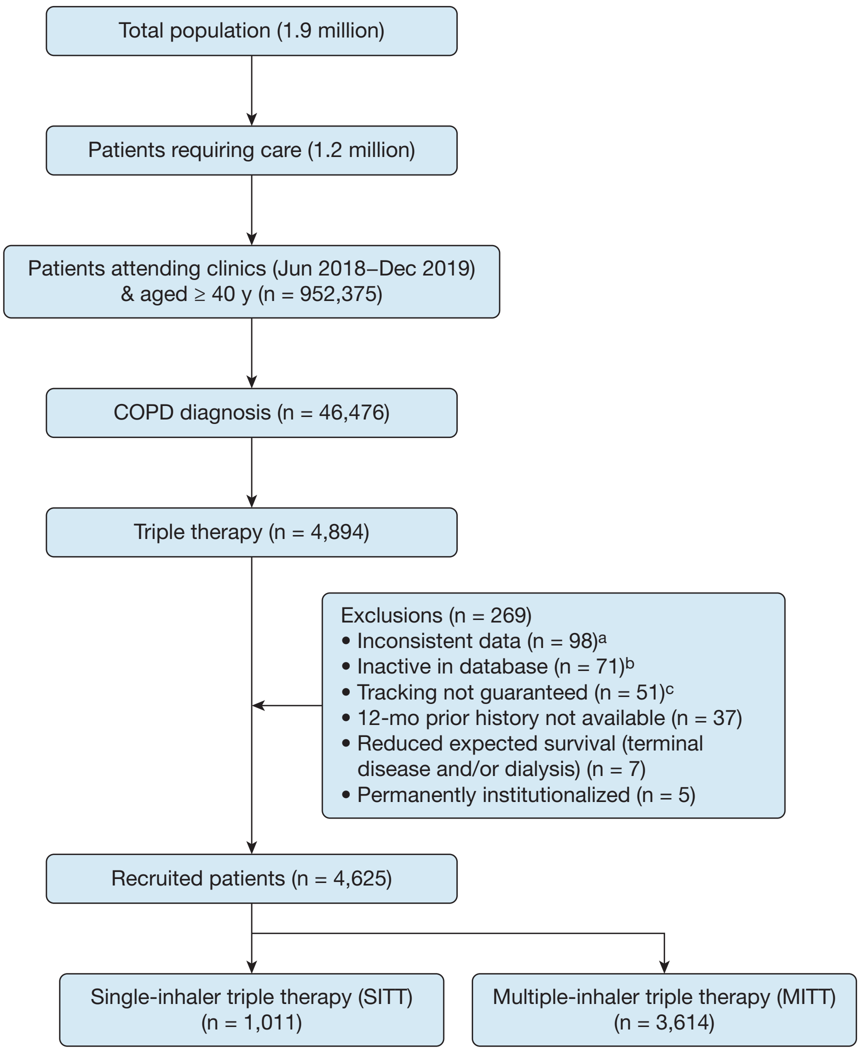

Figure 1 – *Patient disposition. [a]After manual review, data were found to be inaccurate (eg, FEV_1 data not available); [b]no available follow-up after the start of triple therapy (defined as at least two entries in the electronic systems); [c]individuals moved or relocated to areas outside the scope of the Big-Pac database.*

COPD diagnosis of 16.6 (7.8) years. Baseline characteristics were similar between cohorts: no significant differences were seen between groups for the primary baseline variables, although statistically significant differences between groups were found for age when stratified into four groups ($P = .037$) (Table 1).

No significant differences were found between the SITT and MITT groups in the mean number of exacerbations during the year before the index date (0.75 vs 0.68), the proportion of patients with ≥1 exacerbation (48.9% vs 47.0%), or the proportion of patients with moderate (38.0% vs 36.0%) or severe (16.8% vs 17.5%) exacerbations. The mean number of historical severe exacerbations before the index date was higher in the SITT cohort (0.37 vs 0.30; $P = .028$) (Table 1).

In the MITT cohort, all patients previously received dual therapy, mainly with combination ICS/LABA (n = 2,094; 57.9%) or combination LABA/LAMA (n = 1,293; 35.8%). Patients in the SITT cohort previously received either MITT (n = 677; 67.0%) mostly with combination ICS/LABA plus LAMA (n = 499; 49.4%) or combination LABA/LAMA plus ICS (n = 160; 15.8%); or dual combination therapy (n = 334; 33.0%), mostly with ICS/LABA (n = 105; 10.4%) or LABA/LAMA (n = 195; 19.3%).

At the index date, patients in the MITT cohort mainly received combination ICS/LABA plus LAMA (n = 2,175; 60.2%) or combination LABA/LAMA plus ICS (n = 1,298; 35.9%). Patients in the SITT cohort at the index date received triple therapy with either beclomethasone/formoterol/glycopyrronium (n = 582; 57.6%) or fluticasone/vilanterol/umeclidinium (n = 429; 42.4%).

In the primary analysis, treatment persistence rates in the SITT vs MITT cohort were higher at both 6 months (80.6% vs 76.7%; $P = .008$) and 12 months (62.4% vs 53.8%; $P < .001$) (Table 2). Patients initiating SITT had higher

TABLE 1] Baseline Characteristics of the Study Population

Study Cohorts	Single-Inhaler Triple Therapy (SITT)	Multiple-Inhaler Triple Therapy (MITT)	Total	P[a]
No. of patients, No. (%)	1,011 (21.9)	3,614 (78.1)	4,625 (100.0)	
Demographics				
Age, y, mean (SD)	71.2 (13.5)	70.9 (14.5)	70.9 (14.3)	.558
Age range, y, No. (%)				
40-44	83 (8.2)	410 (11.3)	493 (10.7)	.037
45-64	372 (36.8)	1,245 (34.4)	1,617 (35.0)	
65-69	270 (26.7)	840 (23.2)	1,110 (24.0)	
≥ 75	286 (28.3)	1,119 (31.0)	1,405 (30.4)	
Male sex, No. (%)	742 (73.4)	2,675 (74.0)	3,417 (73.9)	.689
BMI, mean (SD)	28.6 (4.4)	28.7 (4.4)	28.7 (4.4)	.424
Comorbidities, No. (%)				
Arterial hypertension	612 (60.5)	2,252 (62.3)	2,864 (61.9)	.303
Dyslipidemia	519 (51.3)	1,846 (51.1)	2,365 (51.1)	.885
Obesity	278 (27.5)	1,021 (28.3)	1,299 (28.1)	.637
Diabetes	274 (27.1)	1,014 (28.1)	1,288 (27.8)	.549
Heart failure	254 (25.1)	878 (24.3)	1,132 (24.5)	.588
Ischemic cardiopathy	182 (18.0)	619 (17.1)	801 (17.3)	.516
Renal failure	130 (12.9)	427 (11.8)	557 (12.0)	.368
Asthma	127 (12.6)	420 (11.6)	547 (11.8)	.413
Depressive syndrome	128 (12.7)	405 (11.2)	533 (11.5)	.201
Malignant neoplasm	126 (12.5)	407 (11.3)	533 (11.5)	.290
Peripheral arterial disease	128 (12.7)	397 (11.0)	525 (11.4)	.138
Stroke	110 (10.9)	405 (11.2)	515 (11.1)	.771
Smoking history, No. (%)				
Person who smokes	246 (24.3)	878 (24.3)	1,124 (24.3)	.980
Person who previously smoked	676 (66.9)	2,458 (68.0)	3,134 (67.8)	.852
No. of chronic conditions, mean (SD)	3.1 (2.0)	3.0 (1.9)	3.0 (1.9)	.511
Charlson Comorbidity Index, mean (SD)	2.3 (1.7)	2.2 (1.6)	2.2 (1.6)	.124
1, No. (%)	438 (43.3)	1,634 (45.2)	2,072 (44.8)	.245
2, No. (%)	259 (25.6)	957 (26.5)	1,216 (26.3)	
≥ 3, No. (%)	314 (31.1)	1,023 (28.3)	1,337 (28.9)	
Severity of airflow limitation, No. (%)				
Moderate	620 (61.3)	2,248 (62.2)	2,868 (62.0)	.614
Severe	374 (37.0)	1,315 (36.4)	1,690 (36.5)	
Very severe	17 (1.7)	51 (1.4)	68 (1.5)	
Time since diagnosis, y, mean (SD)	17.0 (8.3)	16.5 (7.7)	16.6 (7.8)	.075
FEV_1, mean (SD)	58.4 (6.3)	57.3 (6.5)	57.6 (6.5)	.078
Eosinophil count, cells/µL, mean (SD)	293 (128.5)	295.4 (124.5)	294.9 (125.4)	.595
Patients with ≥ 300 eosinophils/µL, No. (%)	268 (26.5)	996 (27.6)	1,264 (27.3)	.507
Exacerbations during previous year				
No. of exacerbations, mean (SD)	0.75 (0.96)	0.68 (0.90)	0.70 (0.91)	.053
Patients with ≥ 1 exacerbation, No. (%)	494 (48.9)	1,698 (47.0)	2,192 (47.4)	.290
0 exacerbations	517 (51.1)	1,916 (53.0)	2,433 (52.6)	.118
1 exacerbation	329 (32.5)	1,214 (33.6)	1,543 (33.4)	
≥ 2 exacerbations	165 (16.3)	484 (13.4)	649 (14.0)	

(Continued)

TABLE 1] *(Continued)*

Study Cohorts	Single-Inhaler Triple Therapy (SITT)	Multiple-Inhaler Triple Therapy (MITT)	Total	P^a
Patients with moderate exacerbations, No. (%)	384 (38.0)	1,300 (36.0)	1,684 (36.4)	.240
No. of moderate exacerbations, mean (SD)	0.38 (0.49)	0.38 (0.48)	0.36 (0.48)	.325
Patients with severe exacerbations, No. (%)	170 (16.8)	631 (17.5)	801 (17.3)	.632
No. of severe exacerbations, mean (SD)	0.37 (0.96)	0.30 (0.84)	0.34 (0.87)	.028

[a]Statistical tests comparing independent groups for qualitative (χ^2) or quantitative (analysis of variance) data or with the nonparametric Mann-Whitney *U* test for quantitative variables that were not normally distributed.

persistence compared with MITT during 12-month follow-up (hazard ratio [HR] = 1.37; 95% CI = 1.22-1.53; $P < .001$) (Fig 2). Mean treatment persistence in SITT and MITT groups was 300.6 and 278.6 days, respectively ($P < .001$) (Table 2). Sensitivity analyses that defined discontinuation as > 90 days without renewing a triple therapy prescription showed comparable results to those for discontinuation for at least 60 days without refill: the HR for > 90 day discontinuation was 1.39 (95% CI = 1.25 to 1.52; $P < .001$).

Although significant differences between SITT and MITT cohorts were found, secondary outcomes (exacerbation, all-cause mortality, and health care resource use and costs) were not adjusted for persistence.

During follow-up, patients in the SITT group had a significantly lower mean number of exacerbations (0.56 vs 0.71; $P < .001$), a lower proportion of patients with one or more exacerbations (38.1% vs 44.4%; $P < .001$), lower proportions of patients with moderate (29.9% vs 33.5%; $P = .031$) or severe exacerbations (11.6% vs 15.4%; $P = .002$), and a prolonged mean time to the first exacerbation (203.3 vs 179.3 days; $P < .001$) compared with the MITT group (Table 3). Patients initiating SITT had a lower risk of exacerbations compared with MITT (HR = 0.68; 95% CI = 0.61-0.77; $P = .001$) (Fig 3).

All-cause mortality rates at 12-month follow-up were 2.9% and 4.4%, respectively (Table 3). During follow-up, patients initiating SITT had a reduced risk of 12-month all-cause mortality compared with MITT (HR = 0.67; 95% CI = 0.63-0.71, $P = .027$) (Fig 4).

Compared with MITT, SITT was associated with significant reductions in the use of COPD-related health care resources and costs. SITT was associated with a reduced mean number of primary care (8.2 vs 10.5; $P <$ .001), specialized care (1.0 vs 1.1; $P = .044$), and emergency room (0.5 vs 0.7; $P < .001$) visits; a reduced proportion of hospitalized patients (11.4% vs 15.4%; $P =$ .001); a reduced mean length of hospitalized stay (2.0 vs 2.6 days; $P = .026$); and reduced use of imaging and other testing, when compared with MITT. Reduced use of health care resources was reflected in adjusted mean annual cost savings (€403) for SITT vs MITT (€2,520

TABLE 2] Treatment Persistence During 12-Month Follow-up

Study Cohort	Single-Inhaler Triple Therapy (SITT) (n = 1,011)	Multiple-Inhaler Triple Therapy (MITT) (n = 3,614)	Total (N = 4,625)	P^a
Persistence, No. (%)				
6 mo	815 (80.6%; 95% CI: 78.2-83.1)	2,772 (76.7%; 95% CI: 75.3-78.1)	3,589 (77.6%; 95% CI: 76.4-78.8)	.008
12 mo	631 (62.4%; 95% CI: 59.4-65.4)	1,944 (53.8%; 95% CI: 52.2-55.4)	2,576 (55.7%; 95% CI: 54.2-57.1)	< .001
Treatment persistence, d, mean (SD)	300.6 (100.9)	278.6 (114.1)	283.4 (111.7)	< .001
Cox proportional hazards model[b]				
HR (95% CI)	1.37 (1.22–1.53)	...	...	< .001

CI = confidence interval; HR = hazard ratio.
[a]Statistical tests comparing independent groups for qualitative (χ^2) or quantitative (analysis of variance) data.
[b]Reference group for Cox proportional hazards model: SITT

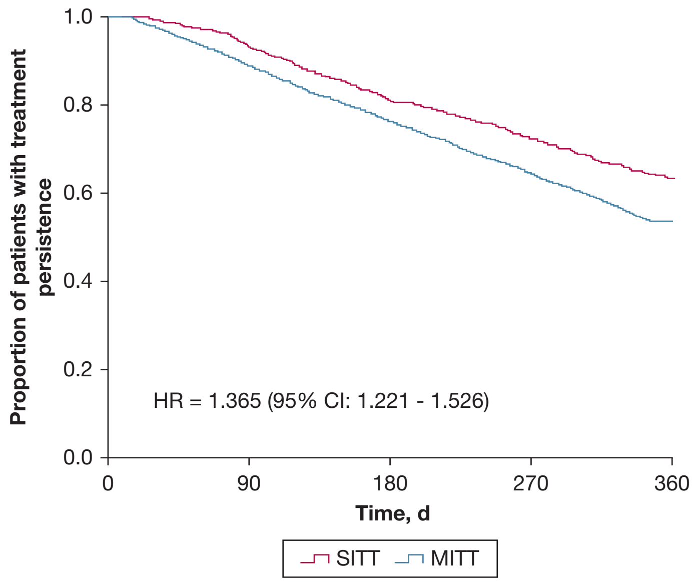

Figure 2 – *Kaplan-Meier graphs for treatment persistence in single-inhaler triple therapy (SITT) and multiple-inhaler triple therapy (MITT). Adjusted for covariates: age, sex, BMI, smoking status, time from diagnosis,* FEV_1*, eosinophil count, heart failure, renal failure, Charlson Comorbidity Index, and previous exacerbations.*

vs €2,923; $P = .006$) (Table 4). Unitary costs are shown in the supplement (e-Table 1).

In both the SITT and MITT groups, persistence was associated with improved exacerbation rates compared with nonpersistence. In the SITT cohort, the mean number of exacerbations in persistent vs nonpersistent patients was 0.04 and 1.44, respectively; and in the MITT cohort it was 0.08 and 1.44, respectively. The proportion of patients with no exacerbations in the SITT group for persistent and nonpersistent patients was 97.5% vs 2.9%, respectively; and in the MITT cohort it was 95.4% vs 9.2%, respectively (Table 5). Mean adjusted health care costs were substantially reduced in persistent patients compared with nonpersistent patients in both SITT (€1,810 vs €3,928) and MITT cohorts (€1,752 vs €4,444), with cost savings of €2,118 and €2,692, respectively (Table 5).

During follow-up, the most common concomitant medications issued were SABA (90.9%), oral corticosteroids (27.4%), and systemic antibiotics (24.3%), and these were given more frequently to the MITT cohort than the SITT cohort: SABA (91.5% vs 88.6%; $P = .005$), oral corticosteroids (28.4% vs 23.5%; $P = .002$), and systemic antibiotics (25.2% vs 21.1%; $P = .007$).

Discussion

Based on data from a national health care database, this observational study demonstrated that COPD patients initiating triple therapy with SITT had increased treatment persistence when compared with MITT patients. This led to a decreased risk of moderate and severe exacerbations, a decreased risk of all-cause mortality, and a reduction in health care resource use. These secondary outcomes were not adjusted for differences in persistence, and the study did not differentiate between different combinations of active agents. Baseline characteristics were similar in each cohort, which precluded propensity score matching. Based on their level of airflow limitation, patients' conditions were classified mainly as moderate or severe. Patients with comorbid asthma (11.8%) were not excluded from the study, which contrasts with most randomized controlled trials for COPD, which have much more stringent inclusion and exclusion criteria compared with effectiveness studies.[25] The study included patients with COPD that were prescribed triple therapy in real-world practice and did not assess the suitability of patients for triple therapy according to guidelines.

At 12-month follow-up, SITT patients had a 37% improvement in persistence compared with MITT patients, leading to a 33% risk reduction in all-cause mortality and a 32% risk reduction in the incidence of exacerbations. Similar improvement in clinically relevant outcomes was reported recently in a European 24-week multicenter, randomized, open-label, phase IV effectiveness study (INTREPID) comparing SITT vs MITT for the delivery of fluticasone furoate/

TABLE 3] Exacerbation Rates and All-Cause Mortality During 12-Month Follow-up

Study Cohort	Single-Inhaler Triple Therapy (SITT) (n = 1,011)	Multiple-Inhaler Triple Therapy (MITT) (n = 3,614)	Total (N = 4,625)	P[a]
No. of exacerbations, mean (SD)	0.56 (0.87)	0.71 (1.00)	0.67 (0.97)	< .001
Patients with exacerbations, No. (%)	385 (38.1)	1,605 (44.4)	1,990 (43.0)	< .001
0 exacerbations	626 (61.9)	2,009 (55.6)	2,635 (57.0)	< .001
1 exacerbation	270 (26.7)	1,043 (28.9)	1,313 (28.4)	
≥ 2 exacerbations	115 (11.4)	562 (15.6)	677 (14.6)	
Patients with moderate exacerbations, No. (%)	302 (29.9)	1,210 (33.5)	1,512 (32.7)	.031
Moderate exacerbations, mean (SD)	0.34 (0.55)	0.39 (0.62)	0.38 (0.6)	.013
Patients with severe exacerbations, No. (%)	117 (11.6)	557 (15.4)	674 (14.5)	.002
Severe exacerbations, mean (SD)	0.22 (0.83)	0.32 (0.81)	0.29 (0.79)	.003
Time until first exacerbation, d, mean (SD)	203.3 (98.5)	179.3 (99.1)	183.9 (99.4)	< .001
Deaths, No. (%)	29 (2.9%; 95% CI: 1.8-3.9)	159 (4.4%; 95% CI: 3.8-5.1)	190 (4.1%; 95% CI: 3.5-4.7)	.027
Time until death, d, mean (SD)	238.8 (88.1)	196.2 (93)	202.8 (93.3)	.023
HR (95% CI) for exacerbations	0.683 (0.607-0.769)	...		.001
HR, (95% CI) for mortality	0.668 (0.625-0.712)	...		.027

[a]Statistical tests comparing independent groups for qualitative (χ^2) or quantitative (analysis of variance) data. Analyses are adjusted for covariates: age, sex, BMI, smoking status, time from diagnosis, FEV_1, eosinophil count, heart failure, renal failure, Charlson Comorbidity Index, and previous exacerbations.

umeclidinium/vilanterol.[20] However, an important point of difference between the current study and INTREPID is that effectiveness was only assessed in terms of symptom control, whereas the impact on exacerbations, adherence, death, or health care resource use was not assessed. In the current study, significant improvement

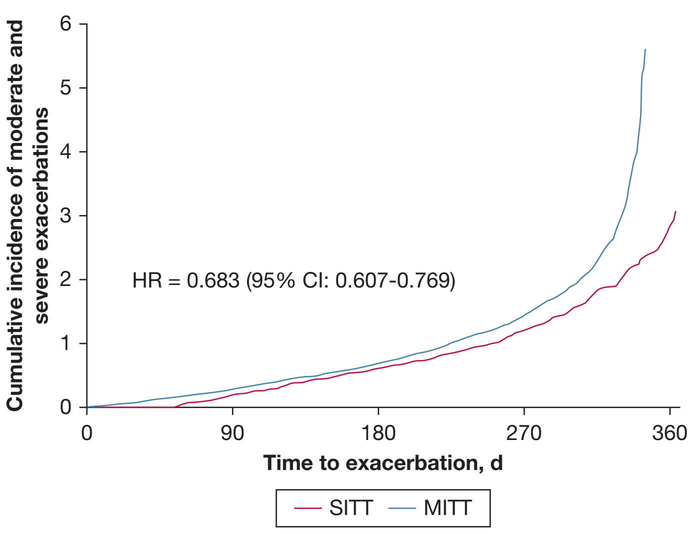

Figure 3 – *Cox proportional hazards model for cumulative incidence of exacerbations in single-inhaler triple therapy (SITT) and multiple-inhaler triple therapy (MITT). Adjusted for covariates: age, sex, BMI, smoking status, time from diagnosis, FEV_1, eosinophil count, heart failure, renal failure, Charlson Comorbidity Index, and previous exacerbations.*

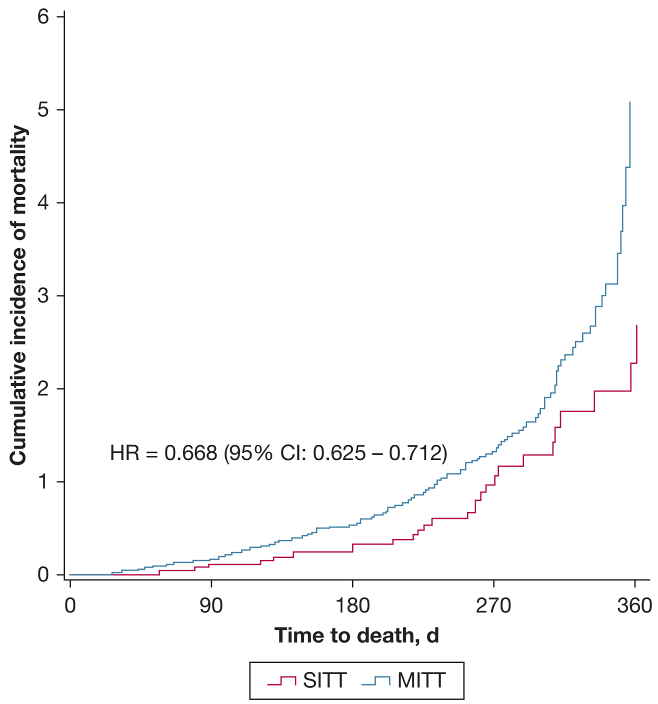

Figure 4 – *Kaplan-Meier graphs for 12-month mortality in single-inhaler triple therapy (SITT) and multiple-inhaler triple therapy (MITT). Adjusted for covariates: age, sex, BMI, smoking status, time from diagnosis, FEV_1, eosinophil count, heart failure, renal failure, Charlson Comorbidity Index, and previous exacerbations.*

in persistence measures in the SITT cohort was accompanied by clinically relevant improvements, including significantly lower 12-month all-cause mortality rate (2.9% vs 4.4%); significantly lower mean number of exacerbations (0.56 vs 0.71); significantly lower proportions of patients with one or more exacerbations (38.1% vs 44.4%), moderate (29.9% vs 33.5%) or severe exacerbations (11.6% vs 15.4%), and a significantly prolonged mean time to the first exacerbation (203.3 vs 179.3 days) compared with the MITT cohort. In other retrospective observational analyses of databases, persistence rates in users of MITT for COPD are generally low, although definitions of persistence were variable.[26-28] Using an allowable treatment gap of no more than 30 days, analysis of a US claims database (IQVIA Adjudicated Health Plan Claims) found that 86% of patients discontinued treatment at 12 months.[26] A UK study, which analyzed electronic health records and hospital episodes in COPD patients, found that the duration of continuous MITT use at 12 months was 23.8%.[27] Finally, in a New Zealand study that evaluated a general practice database, mean persistence (defined as without a gap in an assumed 90-day supply) for COPD patients who initiated MITT was 47.3% at 12 months' follow-up.[28] Recent analysis of a US claims database (IQVIA) found that patients with COPD initiating a defined SITT regimen (fluticasone furoate/umeclidinium/vilanterol) had better adherence and persistence compared with patients initiating MITT.[23]

In our analysis, SITT was also associated with significant reductions in the use of health care resources, with consequent mean annual cost savings of €403 compared with MITT. Similarly, a systematic review, which compared the use of single vs multiple inhalers to deliver the same medications for patients with COPD or asthma, reported that single-inhaler use was associated with reduced health care resource utilization and improved cost-effectiveness.[22] In both the SITT and MITT groups in the current study, persistence was associated with improved exacerbation rates compared with nonpersistence, and substantial adjusted mean annual health care cost savings of €2,115 and €2,700, respectively.

Compared with patients initiating MITT, SITT initiators had a lower usage of concomitant medications such as

TABLE 4] Use of COPD-Related Health Care Resources and Costs During the 12-Month Follow-up

Study Cohort	Single-Inhaler Triple Therapy (SITT) (n = 1,011)	Multiple-Inhaler Triple Therapy (MITT) (n = 3,614)	Total (N = 4,625)	P[a]
Health care resources[b]				
Primary care visits, mean (SD)	8.2 (9.1)	10.5 (10.8)	10 (10.5)	< .001
Specialized care visits, mean (SD)	1.0 (1.3)	1.1 (1.5)	1.0 (1.5)	.044
Emergency room visits, mean (SD)	0.5 (0.8)	0.7 (0.9)	0.6 (0.9)	< .001
Hospitalized patients, No. (%)	115 (11.4)	557 (15.4)	671 (14.5)	.001
No. of hospitalizations, No. (%)				
1	69 (6.8)	408 (11.3)	477 (10.3)	.001
2	32 (3.2)	102 (2.8)	134 (2.9)	
≥ 3	14 (1.4)	47 (1.3)	61 (1.3)	
Hospitalization, d, mean (SD)	2.0 (6.5)	2.6 (7.4)	2.5 (7.2)	.026
Laboratory tests, mean (SD)	1.5 (2.0)	1.5 (2.0)	1.5 (2.0)	.523
Conventional radiology, mean (SD)	0.3 (0.6)	0.4 (0.6)	0.3 (0.6)	.014
Axial tomography, mean (SD)	0.2 (0.7)	0.3 (0.8)	0.3 (0.8)	.004
Magnetic nuclear resonance, mean (SD)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	.055
Other testing, mean (SD)	2.3 (0.5)	2.4 (0.5)	2.4 (0.5)	.003
Costs,[b] €, mean (SD)				
Primary care visits	191 (211.5)	243.5 (251.0)	232 (243.9)	< .001
Specialized care visits	88.5 (119.9)	98.4 (141.8)	96.2 (137.3)	.044
Emergency room visits	62.1 (93.8)	76.9 (104.8)	73.7 (102.7)	< .001
Hospitalization	850.5 (2732.5)	1,090.7 (3,099.8)	1,038.2 (3,024.6)	.026
Laboratory tests	32.8 (43.6)	33.9 (44.9)	33.6 (44.6)	.523
Conventional radiology	5.7 (10.5)	6.6 (11.5)	6.4 (11.3)	.014
Axial tomography	21.3 (64.8)	28.9 (75.9)	27.2 (73.6)	.004
Magnetic nuclear resonance	18.4 (54.0)	22.3 (58.8)	21.5 (57.8)	.055
Other testing	86.7 (19.8)	88.8 (19.1)	88.3 (19.3)	.003
Medication[c]	1,184.9 (1,563.9)	1,253.4 (1,611.9)	1,238.4 (1,601.6)	.229
Healthcare costs, €, mean (SD)	2,541.9 (3,317.6)	2,943.3 (3,785.8)	2,855.6 (3,691.9)	.002
Adjusted health care costs, €[d], (95% CI)	2,520 (2,165-2,874)	2,923 (2,719-3,126)	Difference: –403	.006

[a]Statistical tests comparing independent groups for qualitative (χ^2) or quantitative (analysis of variance) data.
[b]Utilization of health care resources and costs shown are per patient. Unitary costs are shown in e-Table 1.
[c]Medications are single-inhaler triple therapy (SITT), multiple-inhaler triple therapy (MITT), and oral corticosteroids, systemic antibiotics, short-acting β agonists (SABA), xanthines, leukotrienes, and biologics.
[d]Analysis of covariance model using bootstrap sampling simulation (1,000 simulation samples). Contrasts are based on pairwise comparisons between estimated marginal means. Costs are adjusted for age, sex, BMI, smoking status, time from diagnosis, FEV_1, eosinophil count, heart failure, renal failure, and Charlson Comorbidity Index.

SABA and oral corticosteroids, which are associated with relevant side effects.

The study is subject to a number of potential limitations. Data from electronic health care records can be subject to variable quality.[29] However, in this study, data were validated by frequency distribution analysis and carefully reviewed to identify possible recording or coding errors. Although a strength of our study is that patient spirometry data (FEV_1) were collected, allowing an assessment of airflow limitation severity, the absence of specific symptom data represents a limitation regarding the complete assessment of symptoms/risk of exacerbations; additionally, because persistence was defined by the number of prescriptions dispensed, inhaler use by patients and inhalation technique could

TABLE 5] Exacerbations and Costs During Follow-up by Treatment Persistence, in Study Cohorts

Study Cohorts	Single-Inhaler Triple Therapy (SITT) (n = 1,011)		Multiple-Inhaler Triple Therapy (MITT) (n = 3,614)	
	Persistent	Nonpersistent	Persistent	Nonpersistent
Patients, No. (%)	631 (62.4)	380 (37.6)	1,944 (54.0)	1,670 (46.0)
Exacerbations, No. (%)				
Patients with exacerbations	16 (2.5)	369 (97.1)	89 (4.6)	1,516 (90.8)
0	615 (97.5)	11 (2.9)	1,855 (95.4)	154 (9.2)
1	11 (1.7)	259 (68.2)	54 (2.8)	989 (59.2)
≥ 2	5 (0.8)	110 (28.9)	35 (1.8)	527 (31.6)
Exacerbations, mean (SD)	0.04 (0.25)	1.44 (0.82)	0.08 (0.42)	1.44 (0.98)
Patients with moderate exacerbations, No. (%)	12 (1.9)	290 (76.3)	75 (3.9)	1,135 (68)
No. of moderate exacerbations, mean (SD)	0.02 (0.17)	0.87 (0.55)	0.05 (0.25)	0.80 (0.67)
Patients with severe exacerbations, No. (%)	5 (0.8)	112 (29.5)	28 (1.4)	529 (31.7)
No. of severe exacerbations, mean (SD)	0.02 (1.0)	0.57 (1.02)	0.03 (0.29)	0.64 (1.06)
Health care costs, €, mean (SD)	1,695 (2,063)	3,948 (4,368)	1,739 (2,013)	4,345 (4,759)
Adjusted health care costs, €[a]	1,810 (95% CI: 1,538-2,082)	3,928 (95% CI: 3,562-4,287)	1,752 (95% CI: 1,572-1,923)	4,444 (95% CI: 4,242-4,648)

[a]Analysis of covariance model using bootstrap sampling simulation (1,000 simulation samples). Contrasts are based on pairwise comparisons between estimated marginal means. Costs are adjusted for age, sex, BMI, smoking status, time from diagnosis, FEV_1, eosinophil count, heart failure, renal failure, and Charlson Comorbidity Index.

not be ascertained. We also acknowledge that, although persistence per se is not directly an outcome of interest to the patient, it is an outcome that can impact other outcomes. Other limitations are that the diagnosis of exacerbations was based on drug claims (moderate exacerbation) and hospital admission or emergency visit (severe exacerbation) and not made by a physician; because the database does not record the cause of death, it was not possible to ascertain deaths due to COPD; although the study reports data from patients within the Spanish National Healthcare System, possible access of some SITT-treated patients to modern therapies cannot be discounted. Finally, potential socioeconomic or insurance differences were not addressed; however, the BIG-PAC database includes the population treated in the public health system in Spain and, because SITT and MITT are funded by this system, access to treatment does not depend on the patient's purchasing power. Moreover, insurance considerations are not applicable because all patients were treated in the public health system.

Interpretation

Patients initiating SITT had a clinically relevant improvement in persistence, leading to reductions in all-cause mortality and in the incidence of exacerbations, compared with patients initiating MITT. SITT was associated with significant reductions in health care resource use, with consequent cost savings.

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Data availability (where applicable): This was a secondary data study using the BIG PAC database. Public access to the database is open.

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